ACC250: Intro to Financial Accounting

Ch11. Shareholders' Equity

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Learning Objectives

- Record **stock issuance** at **par value** with **additional paid-in capital**.
- Distinguish **authorized**, **issued**, **treasury**, and **outstanding** shares.
- Account for **cash dividends** and **treasury stock**.
- Compare **equity** and **debt** financing.
- Compute **ROE**, **EPS**, and the **P/E ratio**.

Table of Contents

1. Corporations
2. Shares
3. Dividends
4. Shareholders' Equity
   - Contributed Capital
   - Treasury Stock
   - Retained Earnings
5. Ratio analysis
   - Return on Equity (ROE)
   - Earnings Per Share (EPS)
   - Price-to-Earnings (PER) Ratio
6. Concluding illustration



Corporations

a business structure that separates ownership from management and provides limited liability to its owners.

- A common form of business organization.
- _____
  - A corporation is distinguished from its owners (shareholders).
  - Shares can be bought and sold, allowing for easy transfer of ownership.
  - A corporation can enter into contracts, sue and be sued, and own property in its own name.
- _____
  - Owners' liabilities are **limited to their investment** in the corporation.
  - This limited liability protects shareholders' personal assets from being used to satisfy the corporation's debts.

EX. You invested \$10K in a corporation. The corporation incurs \$1M in debts. In case of bankruptcy, how much can you lose?

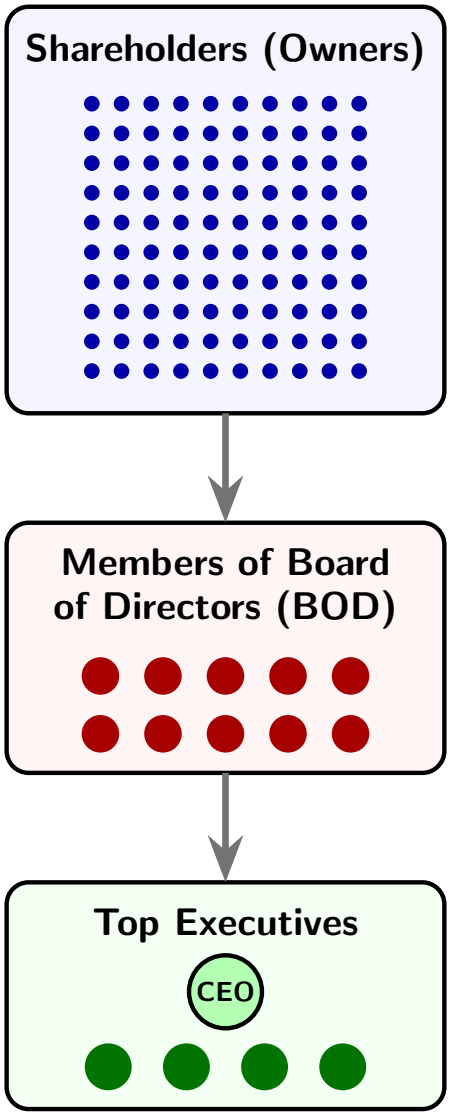
- _____ investment.
- Your personal assets are _____ from the corporation's debts.
- This is the essence of _____.

EX. Suppose a corporation incurs debts of \$1M (including Wages Payable (employees), Accounts payable (suppliers), Notes payable (banks), etc.) but only has assets worth \$600K because of continuing losses. In case of liquidiation:

- How much would the corporation pay to its creditors (employees, etc.)?
  _____, not _____.
- How much can shareholders take from the corporation?
  _____
- But still, the shareholders are not personally responsible for the remaining _____.
- Wait! how much SHE for the firm without liquidation?[1]
  _____

[1] Yes, it can be negative (e.g., American Airlines due to continuing losses in the past)!

Corporate Structure



Vote only on major corporate matters (e.g., election of the BOD; M&A).

Involved in important decision-making (incl. CEO appointment/firing).

Responsible for the day-to-day operations.

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2. Shares
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   - Retained Earnings
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   - Earnings Per Share (EPS)
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Shares[3]

- Corporations issue **shares** to shareholders and receive _____ in return.
- **Shares (a.k.a. stocks)** represent _____; a claim on the corporation's **net assets**[2] and **earnings**.
- If you own 1% of **the shares**, you own _____ of **the corporation**.
- Shares can be bought and sold, allowing investors to **trade ownership**.

[2] Net assets = Assets - Liabilities.

[3] There are more than one type of shares/stocks. We focus on **Common Stock** in this course. Other types include Preferred Stock.

Shares - Common stocks

Stockholders have:

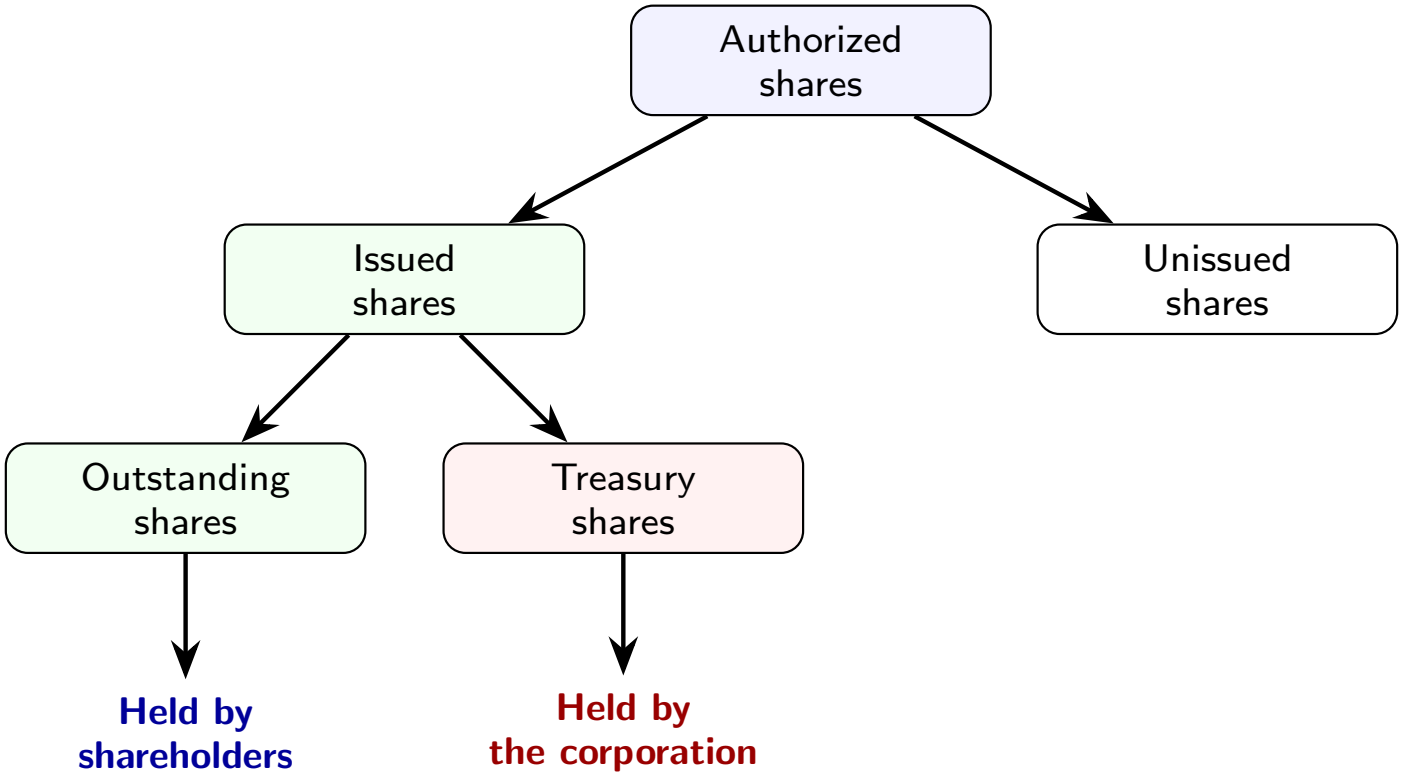
1. **Voting rights** on important matters (e.g., election of BOD members; M&A).
2. **Right to receive dividends**, if any.[4]
3. **Claims to net assets** in case of liquidation (a.k.a. **residual claims**).

[4] Dividends are **not guaranteed**: the board of directors decides whether to declare them.

EX. A corporation has 1K shares of common stock outstanding and you own 100 shares. The corporation has $1M in assets and $800K in liabilities. In case of liquidation,

- $800K would be used to pay off the liabilities.
- The remaining $200K (residual claims) would be distributed to the shareholders.
- You would be entitled to:

Shares - Overview



Shares - Authorized

- Shares must be **authorized** for issuance by the **BOD** before they are actually issued.

EX. The BOD authorized 1M shares for issuance. How much cash is coming in?

__ — authorization alone is **not** a sale of shares.

Shares - Issued

- **Issued shares**: The number of shares that have been **sold to investors**.
- **Unissued shares**: Authorized shares **minus** Issued shares.

EX. A corporation is authorized to issue 1M shares and has sold 700K shares to investors.

Type of shares	Number of shares
Authorized shares	
Issued shares	
Unissued shares	

Shares - Repurchased (= Treasury Stock)

- Firms can **repurchase** their own stocks from shareholders.
- It reduces the shares that are **available for trading** in the market.
- The repurchased stocks are **held by the company**: **Treasury Shares**.

EX. A corporation has authorized 1M shares, issued 700K shares, and repurchased 100K shares as treasury stock.

Type of shares	Number of shares
Authorized shares	
Issued shares	
Treasury shares	

Shares - Outstanding

- **Outstanding shares**: Shares that are currently held by investors.

EX. The same corporation has issued 700K shares and holds 100K shares as treasury stock. How many shares are **outstanding**?

_____ shares (= 700K - 100K).

Shares - Comprehensive Example

EX. The balance sheet for Crutcher Corporation reported 200K shares outstanding, 300K shares authorized, and 20K shares in treasury stock. Compute the **maximum number of new shares** that Crutcher could issue.

Type of shares	**Number of shares**
Authorized shares	300,000
Outstanding shares	200,000
Treasury shares	20,000
Issued shares (Outstanding + Treasury)	_____
Shares available for issuance (Authorized - Issued)	_____

1 Corporations

2 Shares

3 Dividends

4 Shareholders' Equity
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5 Ratio analysis
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6 Concluding illustration

Dividends

- A distribution of a portion of a corporation's _____ to its shareholders.
- Firms usually pay dividends when they have **excess cash** and **stable earnings**.
- _____.
- Cash payment, though _____ an Expense![5]
- Dividends reduce _____.

Retained Earnings Equation

End. RE = Beg. RE + Net Income - Dividends

[5] Recall: Expenses are those that are incurred in the process of generating revenue. Does Dividends fit this definition?

Dividends

- Dividends are paid **only to shareholders**; _____ for treasury stocks (held by the corporation itself).

EX. Netpass Company has 300K shares of common stock authorized, 270K shares issued, and 100K shares of treasury stock. The board of directors declares a dividend of \$1 per share of common stock. **How much is the total dividend?**

Item	Amount
Issued shares	270,000
Treasury shares	(100,000)
Outstanding shares	_____
Dividend per share	\$1
Total dividends	_____

1 Corporations

2 Shares

3 Dividends

4 Shareholders' Equity
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- Earnings Per Share (EPS)
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6 Concluding illustration

Shareholders' Equity

The residual interest in the assets of the corporation after deducting liabilities. It represents the ownership interest of shareholders in the corporation.

- **Contributed capital (C.C)**
- **Treasury stock (T.S)** (a.k.a. stock repurchase; stock buyback)
- **Retained earnings (R.E)**
- Accumulated other comprehensive income[6]

SHE= C.C - T.S + R.E

[6] covered in ACC301 and ACC302.

Contributed Capital

Contributed Capital

The amount of capital that shareholders have contributed to the corporation.

- Main components
  - Common stock
  - Additional paid-in capital (APIC)
  - Preferred stock[7]

[7] covered in ACC301 and ACC302.

Contributed Capital

Common Stock, Par Value (Equity)

The nominal value assigned to each share of stock.

- It used to be a legal requirement
- But now, it is often set at a very low amount (e.g., \$0.01).
- This is not the **market value** of the stock.

Additional Paid-in Capital (APIC, Equity)

The amount received from shareholders in excess of the par value of the stock.

EX. A corporation issues 1K shares of common stock with a par value of \$1 per share at a price of \$10 per share.

EX. A corporation issues 2K shares of common stock with a par value of \$2 per share at a price of \$15 per share.

The distiction between Par Value and APIC is important **just for accounting purposes**.

Treasury Stock

Treasury Stock (Contra Equity)

The cost value of shares repurchased by the corporation.

- Shares **repurchased** by the firm; **held** by the firm.
- Share repurchases
  - ⇒ _____.
  - ⇒ Net assets (i.e., SHE = A - L) _____.[8]
  - ⇒ **Treasury shares reduce SHE!**
- Firms are more likely to buy back when the stock price is _____.

[8] Understand it this way: the company is being well managed, so it can do the same with fewer resources (i.e., lower cash); the company can repurchase its own stocks (i.e., pay cash to some investors by buying their shares back). So lower cash; lower assets; lower SHE!

Retained Earnings

Retained Earnings (Earned Capital, Equity)

The cumulative amount of net income retained in the business after dividends are paid.

- Retained earnings (RE) equation:

$$\text{Ending RE} = \text{Beginning RE} + \text{NI} - \text{Div}$$

- Net income (NI): Increases retained earnings.
- Dividends (Div): Decrease retained earnings.

EX. A corporation has beginning retained earnings of $50,000, net income of $20,000, and dividends of $5K.

Item	Amount ($)
Beginning Retained Earnings	_____
Add: Net Income	_____
Less: Dividends	_____
Ending Retained Earnings	_____

Table: Statement of Retained Earnings

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Return on Equity (ROE)

Return on Equity (ROE)

Return (net income) on \$1 of equity investment.
Calculated as: $\frac{\text{Net Income}}{\text{Average Shareholders' Equity}}$ if Preferred Dividends$= 0$.

- Measures how much, for \$1 of equity investment, the firm generates in net income.
- Higher ROE values generally indicate better performance.

EX. A corporation has net income of \$100K and average shareholders' equity of \$500K.

- ROE = \$100K / \$500K = 0.20 or 20%
- This means the corporation generated a return of 20% on its equity investment.

Earnings Per Shares (EPS)

Earnings Per Share (EPS)

A measure of a company's profitability on a per-share basis.
Calculated as: $\frac{\text{Net Income}}{\text{Average Shares Outstanding}}$ if Preferred Dividends$= 0$.

- Measures how much, for each share of common stock, the corporation generates in net income.
- Higher EPS values generally indicate better performance.

EX. A corporation has net income of \$100K and average shares outstanding of 50K.

- EPS = \$100K / 50K = \$2.00
- This means the corporation generated \$2.00 in earnings for each share of common stock outstanding.

Treasury Stock, ROE, and EPS

- Treasury Stock ⇒ _____ Shares Outstanding ⇒ _____ EPS
- Treasury Stock ⇒ _____ SHE ⇒ _____ ROE

EX. A firm has net income of $100K and average shares outstanding of 50K.

- How much is the EPS without stock repurchases
  - EPS = $100K / 50K = _____
- How much is the EPS if the corporation repurchases 10K shares as treasury stock?
  - EPS = $100K / (50K - 10K) = _____

⇒ Purchasing its own stock can **boost the firm's financial performance metrics**!

Price-to-Earnings (P/E) Ratio

Which one is cheaper? (assuming that they have the same growth potential)

Q1. Same EPS with different Price

- A firm that generates $10 EPS; its stock price is now $100.
- A firm that generates $10 EPS; its stock price is now $50. → _____

Q2. Same Price with different EPS

- A firm that generates $10 EPS; its stock price is now $100.
- A firm that generates $20 EPS; its stock price is now $100. → _____

Q3. Different EPS and Price

- A firm that generates $10 EPS; its stock price is now $100.
- A firm that generates $5 EPS; its stock price is now $20. → _____

⇒ The **P/E ratio** can answer this!

Price-to-Earnings (P/E) Ratio

Price-to-Earnings (P/E) Ratio

A valuation ratio that compares a company's current share price to its EPS.
Calculated as: $\frac{\text{Stock Price per Share}}{\text{Earnings Per Share (EPS)}}$

- Measures how much investors are willing to pay for each dollar of earnings.
- Higher P/E ratios indicate currently higher prices relative to earnings, i.e., Premium!
- Why?
  1. the stock is overvalued.
  2. investors expect that future earnings will be higher.

Comparison of United Airlines and Delta Airlines (**End of Nov 2025**)

Metric	**UAL**	**DAL**
Stock Price	\$92.20	\$58.57
2025 EPS forecast from CNBC	\$10.82	\$6.02
P/E	8.52	9.73

- Assume they have similar growth prospects. Which one is undervalued?

_____.

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B/S example 1: Growing Retained earnings

- Year 1: Raised $1M in capital
- Year 1: Borrowed $2M from a bank
- Year 1: Earned $200K in net income.
- Year 2: Earned $300K in net income.
- Year 3: Earned $400K in net income.
- **Calculate Retained Earnings (RE) at the end of each year.**[9]

Balance Sheet	Year 1	Year 2	Year 3
Assets			
...	...	...	...
Total Assets	_____	_____	_____
Liabilities and Shareholders' Equity			
Liabilities			
...	...	...	...
Total Liabilities	2,000,000	2,000,000	2,000,000
Shareholders' Equity			
Contributed Capital (C.C)	1,000,000	1,000,000	1,000,000
Retained Earnings (R.E)	_____	_____	_____
Total Shareholders' Equity	_____	_____	_____
Total Liabilities and Shareholders' Equity	_____	_____	_____

[9] The below B/S can be made from the information above.

B/S example 2: Growing Retained Earnings even with dividends

- Year 1: Raised $1M in capital
- Year 1: Borrowed $2M from a bank
- Year 1: Earned $200K in net income.
- Year 2: Earned $300K in net income.
- Year 3: Earned $400K in net income.
- Year 3: **Paid $250K dividends**.
- **Calculate Retained Earnings (RE) at the end of each year.**[10]

Balance Sheet	Year 1	Year 2	Year 3
Assets			
...	...	...	...
Total Assets	3,200,000	3,500,000	_____
Liabilities and Shareholders' Equity			
Liabilities			
...	...	...	...
Total Liabilities	2,000,000	2,000,000	_____
Shareholders' Equity			
Contributed Capital (C.C)	1,000,000	1,000,000	_____
Retained Earnings (R.E)	200,000	500,000	_____
Total Shareholders' Equity	1,200,000	1,500,000	_____
Total Liabilities and Shareholders' Equity	3,200,000	3,500,000	_____

[10] The below B/S can be made from the information above.

B/S example 3: Growing Retained Earnings even with dividends

- Year 1: Raised $1M in capital.
- Year 1: Borrowed $2M from a bank.
- Year 1: Reported **$600K losses**.
- Year 2: Reported **$800K losses**.
- **Calculate Retained Earnings (RE) at the end of each year.**[11]

Balance Sheet	Year 1	Year 2
Assets		
...	...	...
Total Assets	2,400,000	
Liabilities and Shareholders' Equity		
Liabilities		
...	...	...
Total Liabilities	2,000,000	
Shareholders' Equity		
Contributed Capital (C.C)	1,000,000	
Retained Earnings (R.E)	(600,000)	
Total Shareholders' Equity	400,000	
Total Liabilities and Shareholders' Equity	2,400,000	

- If Retained Earnings is negative, it is called **accumulated deficit**.
- The company has an **accumulated deficit** of **(400K)**.

[11] The below B/S can be made from the information above.